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IARCO RESEARCH PROPOSAL

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Research Topic: HOLONET: Integration of a Portable Sensor & AI in Digital Holographic Imaging Through Neural Network & Polarization For Real-Time Classification & Detection of Microplastics

Introduction

Indonesia has one of the world's longest coastlines, boasting the highest diversity of corals and mangroves [1][3][5]. This therefore benefits the nation by sustaining its economy, prosperity, and resilience in measurable ways. Aquatic ecosystems benefit mangroves and reefs, which protect coastal communities from disasters, therefore reducing the impact of tsunamis and floods [2][15]. Marine and coastal ecosystems help Indonesia meet its climate change commitments, which not only regulate the Earth's climate but also provide a home to thousands of coral and fish species—therefore providing benefits towards the nation's economy, prosperity, and food security within society [1][3].

However, plastic waste has gradually become one of the most important issues that significantly affect the Indonesian aquatic and freshwater ecosystem. Eventually, the abandonment of plastic waste in the environment is naturally fragmented into smaller sizes due to oxidation, temperature, ultraviolet radiation (UV), and physical pressure, resulting in emerging pollutants, so-called microplastics (MP) with a size ranging from 1 µm to 5 mm [6][16][18].

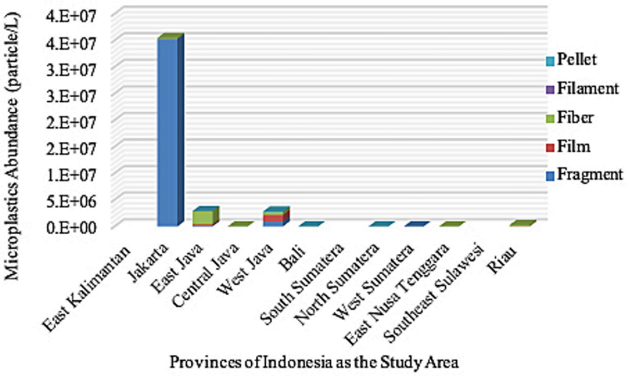


Fig. 1. Microplastic Abundance (MP) in Indonesian waters. Data adapted from [4]

Analysis of 2021 data on MP abundance shows a distinct difference in Jakarta City compared with all other provinces in Indonesia [7]. Following this, the highest microplastic abundance was in Jakarta City, due to the water continuously having higher MP abundance based on the data, which consists of higher fragments [7][12]. Meanwhile, other provinces, such as East Java, Central Java, West Java, Bali, and North Sumatra, account for smaller shares of MP abundance [7]. River-level studies confirm that this contamination extends across the archipelago, from the Cisadane and Cimandiri rivers in West Java [11][14] to the Kahayan River in Kalimantan [13] and the coastal waters of Bandar Lampung [8]. Overall, the data clearly emphasize that Indonesian waters are present through rivers and nearby coastal seas; Jakarta has been affected in the longer term in MP abundance, which hasn't been resolved until today [12].

Therefore, to address these such gaps, this study focuses on developing **HOLONET**, an integration of a portable sensor and AI (Artificial Intelligence) in digital holographic imaging through a neural network and polarization for real-time classification and detection of microplastics in the water.

Research Questions

The study of HOLONET, an integration of a portable sensor and AI (Artificial Intelligence) in digital holographic imaging through a neural network and polarization, that transmits for real-time classification and detection of microplastics in the water, and generates reports within the application. Specifically, this study intends to answer the following questions:

1. How effective is the Holonet device in terms of: **a.** Range of Area Coverage in Water using LoRa-GSM **b.** Speed and Accuracy of CMOS-based imaging sensor **c.** Tracking Exact Location **d.** Detection Accuracy of Particle size, shape, classification, environmental variables, and microplastic abundance

2. How effective is the HoloNet app in terms of: **a.** Receiving Alert Signals **b.** Accuracy Analysis in AI-polarization-based results **c.** GPS tracking and mapping **d.** Alert and Report Management

3. Is there a significant difference in the HoloNet device in terms of: **a.** Speed of particle, , pH, and turbidity detection **b.** Time interval of sending of alert signals **c.** Tracking of exact location and mapping

4. Is there a significant difference in the HoloNet app in terms of: **a.** Generating AI analysis reports accuracy and managing Alerts **b.** Display performance of real-time detection of particle sizes and microplastic abundance

5. What is the rate of acceptability of HoloNet in terms of: **a.** Functionality **b.** Durability **c.** Accessibility **d.** Reliability

Literature Review

From a global perspective, microplastics are widely defined as plastic fragments between 1 nanometer and 5 millimeters in width [6][9]. Small pieces, however, have spread more quickly through distribution in several ways. Over time, they eventually break down into microplastics from discarded plastic products such as water bottles and cling wrap, as synthetic fabrics like polyester gradually shed microplastic fibers when being washed; this fragmentation is driven primarily by oxidation under UV exposure, together with heat and mechanical stress [9][16][18]. The potential harm to the human body is increasingly documented and is linked to exposure through contaminated aquatic sources, with microplastics found in fish and bivalves consumed by coastal communities [7][8][13][9].

In Indonesia, microplastic abundance has severely affected the water environment over time, especially in cities with high microplastic abundance like Jakarta, which can hold microplastics due to the increasing population yet improper management of plastic disposal, which therefore accelerates the distribution and spread of microplastics within waters, rivers, lakes, and even air [7][10]. A researcher discovered that microplastics began to spread into the rain—which has long been a symbol of freshness and purity is no longer completely clean [10]. It was found that rainwater in Jakarta contains dangerous microplastic particles from human activities in urban areas; moreover, it was stated that microplastics originate from human activities in urban areas [10]. Therefore, this causes the presence of microplastics in every rainwater sample taken in the capital; over time, these small fragments of microplastics eventually affect socioeconomic status and environmental state severely in these places [7][10].

That being said, researchers responded by conducting technological work and research in order to address the inevitable presence of microplastics within certain areas, by training machine learning to predict the abundance of microplastics in rivers and other environments. Such an example of research conducted in 2025 demonstrates the estimation of the quantity of microplastics across 24 provinces in Indonesia—with analysis that utilized parameter targets of microplastic counts and environmental factors and anthropogenic factors as feature parameters [17]. The study shows a coefficient of determination (R^2) of 0.838 and a mean absolute percentage error (MAPE) of 0.242 on unseen testing data from the Tree algorithm's performance, thereby highlighting strong predictive capability. However, the study uses machine learning predictive modeling based on macro-environmental parameters, not real-time physical measurement of actual microplastic particles in water samples.

Therefore, to overcome these limitations, this study introduces HOLONET, an integrated system combining digital holographic imaging, polarization analysis, and neural networks. By directly and in real time detecting on-site optical measurements and physically detecting microplastics in water samples, this includes multi-parameter morphological classification that leverages digital holography to classify microplastics by size, shape, and physical characteristics, and application interfaces that support communication and display with users for fixation on microplastic detections as well as offer immediate pollution data. Overall, the prior work highlights the real-world challenges of HOLONET that must address real-time polarization for immediate signals, displaying classification and detection of microplastics in diverse water samples in Indonesia.

Proposed Methodology: This method is a quantitative approach.

1. **Hardware Development:** This section consists of a diode laser module (532 nm green or 635 nm red) with adjustable power to produce the coherent illumination required for holographic interference. The polarization filters contain linear polarizers and Quarter-Wave Plates that are placed in the optical path to analyze the polarization dynamics and birefringence of microplastic particles, as well as non-polarizing beam-splitter cubes and front-surface mirrors to divide and recombine the reference and object beams, or a direct inline holographic setup; a high-speed CMOS image sensor will also be installed within the hardware for higher-resolution monochrome or color imaging without an infrared filter. Meanwhile, a quartz or optical glass flow cell will be placed in a microfluidic chamber through which the water sample passes under laser illumination, with a micro-diaphragm pump to stabilize the pulse-free flow rate through the sensing volume; however, to prevent oversized debris from clogging the silicone tubing, barbed connectors will be in line with micro-filters within the water samples.
2. **Software Development:** The HoloNet Application is built in the programming language Flutter (Dart), which works seamlessly on Android and iOS with built-in support for mapping (Google Maps API) and push notifications. For displaying real-time microplastic abundance, receiving alert signals, GPS mapping, and generating necessary AI analysis reports, it requires integration of PyTorch for customized neural network architectures. Lastly, the backend framework can be done with FastAPI (Python) for high-performance and native support for communication with IoT devices, as well as a

database for quick real-time synchronization between the HoloNet sensor and mobile app.

3. **Artificial Intelligence Development:** Meanwhile, an AI model is trained using YOLO, OpenCV, and NumPy for image preprocessing, using reconstruction algorithms from holographic interference patterns and polarization in microplastic detection. The detection method utilized patch-based analysis to evaluate localized image patches or segment independently rather than assessing the whole image globally.

4. **Testing & Evaluation:** Testing and evaluation begin with hardware benchmarking, evaluating laser coherence, flow cell stability, and camera capture rates. As well as running the calibration beads (known sizes like 10 µm, 50 µm, 100 µm) through the microfluid flow cell to establish the baseline resolution and accuracy. Meanwhile, test the YOLO or PyTorch model using a split dataset of 80% training, 10% validation, and 10% testing, utilizing diverse microplastic types such as fibers, fragments, and pellets. The performance evaluation will be measured according to the standard metrics of machine learning. Meanwhile, the software must be evaluated for measuring the end-to-end latency from image capture at the optical sensor, fast communication database transmission (FastAPI/IoT), and final UI alert display on the application. Therefore, this testing and evaluation will be conducted through urban rivers in Jakarta and coastal waters, with subsequent accumulation of water samples across varying levels of turbidity, pH, and salinity to evaluate real-world sensing robustness.
5. **Acceptability & Rating:** The orange troops (Environment Office) and blue troops (Water Resources Agency) field personnel, along with environmental researchers and local agency officials, will evaluate the HoloNet system to determine its practical usability and field performance. Meanwhile, the evaluation will be conducted using a structured Likert-scale questionnaire (ranging from 1 = Strongly Disagree to 5 = Strongly Agree) to assess four core dimensions of functionality, durability, accessibility, and reliability of the system.
6. **Performance Evaluation & Recommendations:** The overall performance evaluation of the HoloNet system will be based on the detection accuracy and AI-image-based analysis accuracy, communication reliability, and user acceptability rating. Therefore, the efficiency and effectiveness of the system will be statistically assessed by the detection and image analysis accuracy (precision, recall, F1-score) and mean Average Precision (mAP) of the YOLO/PyTorch models, and the system and communication reliability of the optical detection through server transmission to its application. The recommendation will focus on empirical findings to refine optical lens alignment, enhance neural network training datasets, and improve hardware waterproofing for long-term use.

Expected Outcome:

The study is expected to demonstrate the effectiveness and efficiency of HOLONET as a portable sensor integrated with AI in a digital holographic system for detecting microbial activity and water quality through polarization in particular rivers in urban areas and aquatic environments. Therefore, the evaluation focuses on the YOLO classification and detection of microplastics, microbial activity in water, stability of LoRa communication in remote areas, and the usability of a mobile application for real-time monitoring and detection through mapping. Results will focus on improving guidance in polarization and detection accuracy systems for enhanced AI-image-based analysis accuracy.

Potential Limitation:

Limitations in this research include varying microbial activity and other water-quality parameters, especially during long-term use of the prototype's waterproof protection; therefore, the system may experience occasional false positives, delayed detection, or AI-analysis errors, including polarization analysis through captured images for necessary detection.

Conclusion:

In conclusion, HOLONET offers a bridge to aggregated water quality monitoring data by analyzing polarization through microbial activities in the utilization of AI in digital holographic imaging through neural networks; this study aims to demonstrate the solution in assisting local sanitation workers and their teams to enhance their efficiency and effectiveness in subsequently deploying to rivers for major clean-up operations and real-time water monitoring and detection in Jakarta urban areas and beyond the aquatic environment that increase environmental cleaning.

Project Facilities:

The project will span approximately about 9 months. The first 2 months involve the assembly of its hardware and devices, including installation of a holographic setup for its polarization analysis dataset. The next 5 months will be focused on training the YOLO model, increased enhancement of mobile app development, and careful field-testing setup for the prototype. Finally, the next 2 months will focus on

analyzing data, finalizing results, and filling out the report.

Post-program plan:

Upon completion of the project research, the researchers will refine the device through additional field validation and present their findings at academic conferences or publish them in a relevant journal. This includes partnerships with local government units, and Jakarta's sanitation workers are also being considered for potential pilot implementation.

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